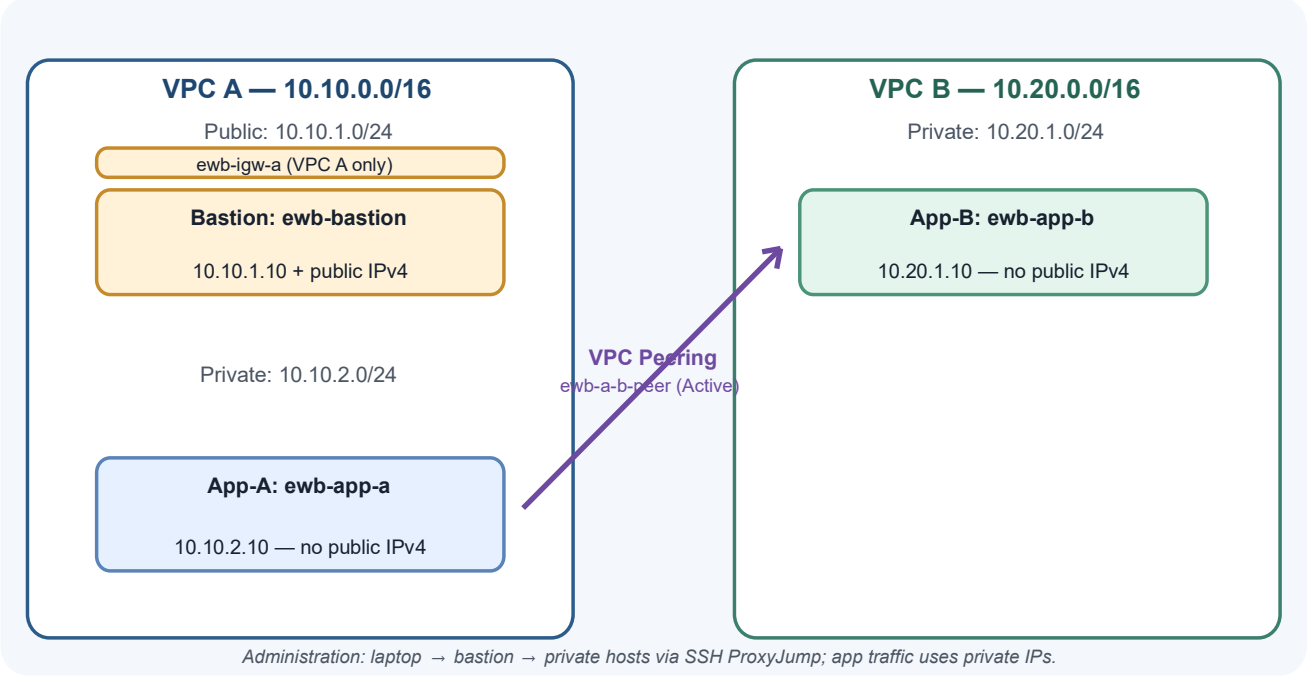
AWS VPC PEERING

EWB Cloud Labs — Student Project Report

Project objective: Connect two non-overlapping Amazon VPCs using VPC peering, enable private IPv4 communication between two private application hosts, and validate access controls using positive and negative tests.

Student name	**Arnak Das Karmakar**
Batch / class	**Aws Internship Batch 1**
AWS Region	Asia Pacific (Mumbai) — ap-south-1
Completion date	29 October 2026
Project	EWB-VPC-Peering



Administration: laptop → bastion → private hosts via SSH ProxyJump; app traffic uses private IPs.

Prepared from the EWB Cloud Labs Student Project Workbook and the test results shared during implementation. Actual AWS resource IDs and console screenshots were not provided in the conversation; fill in the marked fields and insert your own screenshots before submission.

1. Introduction

This project demonstrates private network connectivity between two Amazon Virtual Private Clouds (VPCs) using VPC peering. VPC A contains a public bastion host and a private application host. VPC B contains a second private application host and has no internet gateway. The application instances communicate using private IPv4 addresses; the bastion is used for administrative SSH access.

2. Objectives

• Create two VPCs with non-overlapping IPv4 CIDR blocks and explicitly associated subnets.

• Configure public and private route tables, an internet gateway for VPC A, and a VPC peering connection.

• Limit inbound SSH and ICMP access to the required source addresses.

- Verify baseline failure, forward and reverse private connectivity, SSH access through the bastion, and controlled ICMPfailure/recovery.
- Document resources and clean up lab resources after evidence has been saved.

3. Architecture and resource plan

The two VPC CIDR ranges do not overlap. The peering connection provides a private route between the VPCs but does not make VPC B use VPC A's internet gateway.

Resource	Name	CIDR / private IP	Notes
VPC A	ewb-vpc-a	10.10.0.0/16	Contains public and private subnets
Public subnet A	ewb-a-public	10.10.1.0/24	Bastion subnet; route to IGW
Private subnet A	ewb-a-private	10.10.2.0/24	App-A; no public IPv4
VPC B	ewb-vpc-b	10.20.0.0/16	No internet gateway
Private subnet B	ewb-b-private	10.20.1.0/24	App-B; public IPv4 auto-assignment disabled
Bastion EC2	ewb-bastion	10.10.1.10 + public IPv4	SSH entry point
Application EC2 A	ewb-app-a	10.10.2.10	Private host
Application EC2 B	ewb-app-b	10.20.1.10	Private host
VPC peering	ewb-a-b-peer	Peering ID: **pcx-03fe402e409d93cae**	Reported Active during implementation

Record before submission: VPC IDs, subnet IDs, route table IDs, EC2 instance IDs and the peering ID (pcx-...). These exact identifiers were not available in the shared test outputs.

4. Route table configuration

Route table	Subnet association	Required routes
ewb-rt-a-public	ewb-a-public	10.10.0.0/16 → local 0.0.0.0/0 → ewb-igw-a 10.20.0.0/16 → peering connection
ewb-rt-a-private	ewb-a-private	10.10.0.0/16 → local 10.20.0.0/16 → peering connection
ewb-rt-b-private	ewb-b-private	10.20.0.0/16 → local 10.10.0.0/16 → peering connection

The forward route directs App-A traffic for 10.20.0.0/16 to the peering connection. The return route in VPC B directs traffic for 10.10.0.0/16 back through the same peering connection. Both directions are necessary for the ping replies to return.

5. Security configuration

Security group	Inbound rule	Source	Purpose
ewb-sg-bastion	SSH / TCP 22	Student current public IPv4 /32	Restrict laptop access to bastion
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32	SSH from bastion
ewb-sg-app-a	Custom ICMP IPv4 Echo Request	10.20.1.10/32	Allow ping initiated from App-B
ewb-sg-app-b	SSH / TCP 22	10.10.1.10/32	SSH from bastion
ewb-sg-app-b	Custom ICMP IPv4 Echo Request	10.10.2.10/32	Allow ping initiated from App-A

The lab uses stateful security groups and default outbound allow-all rules as specified in the workbook. SSH and ICMP are not opened to 0.0.0.0/0. App-A and App-B must have no public IPv4 address.

6. Implementation and validation

The VPC peering connection ewb-a-b-peer was reported as Active. The following test results were shared during the project workflow.

Test	Command / action	Observed result	Status
Baseline before peering/routes	App-A: ping -c 4 10.20.1.10	4 transmitted, 0 received, 100% packet loss	PASS — expected failure
Forward path	App-A: ping -c 4 10.20.1.10	4 transmitted, 4 received, 0% packet loss	PASS
Reverse path	App-B: ping -c 4 10.10.2.10	4 transmitted, 4 received, 0% packet loss	PASS
Private SSH	Laptop: ssh ewb-app-b	User reported connection to App-B shell	PASS — capture screenshot
Controlled failure	Remove only App-B ICMP Echo Request source 10.10.2.10/32; ping from App-A	4 transmitted, 0 received, 100% packet loss	PASS
Recovery	Restore App-B ICMP rule; repeat ping from App-A	Four echo replies were shown after restoration; final summary was not included in the last message	PASS — capture final summary

Forward and return path explanation

For a ping from App-A (10.10.2.10) to App-B (10.20.1.10), the App-A subnet route table sends 10.20.0.0/16 to the peering connection. VPC B's route table sends return traffic for 10.10.0.0/16 back through peering. The destination security group allows the ICMP Echo Request from the specific source host. Security groups are stateful, allowing replies to permitted traffic.

For administrative access, the laptop connects to the public bastion over SSH. OpenSSH ProxyJump then creates a separate SSH connection from the bastion to the private destination. The private application hosts do not need public IP addresses.

7. Results and discussion

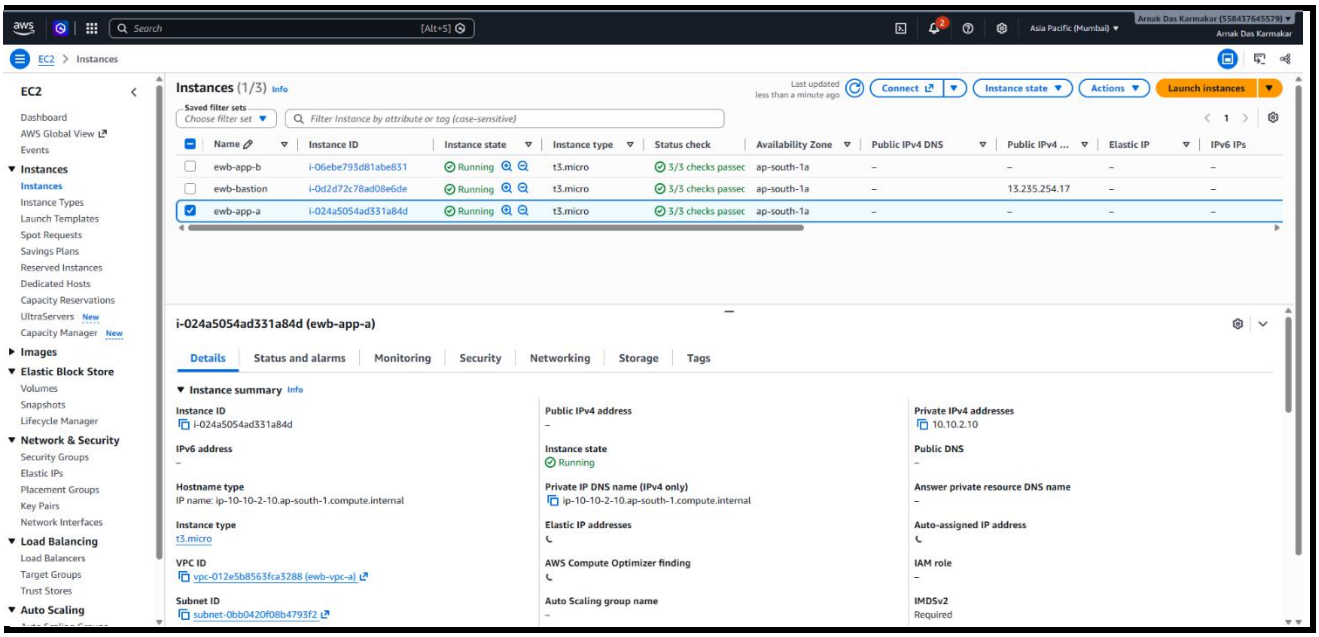
The observed ping tests demonstrate private IPv4 connectivity between the two application hosts in both directions after the peering routes were configured. The baseline test failed before connectivity was enabled, as expected. Removing the

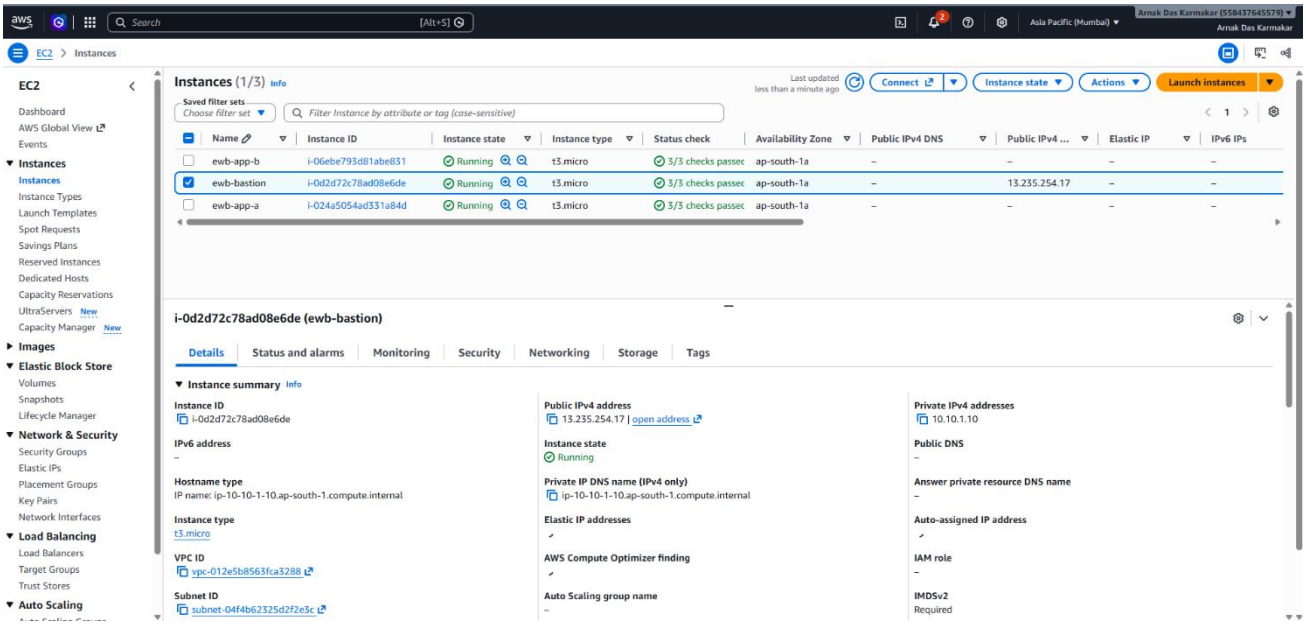
specified App-B ICMP inbound rule stopped replies from App-A, and restoring the rule allowed replies to resume. This controlled test demonstrates that the route path alone is not sufficient; security-group policy must also permit the traffic.

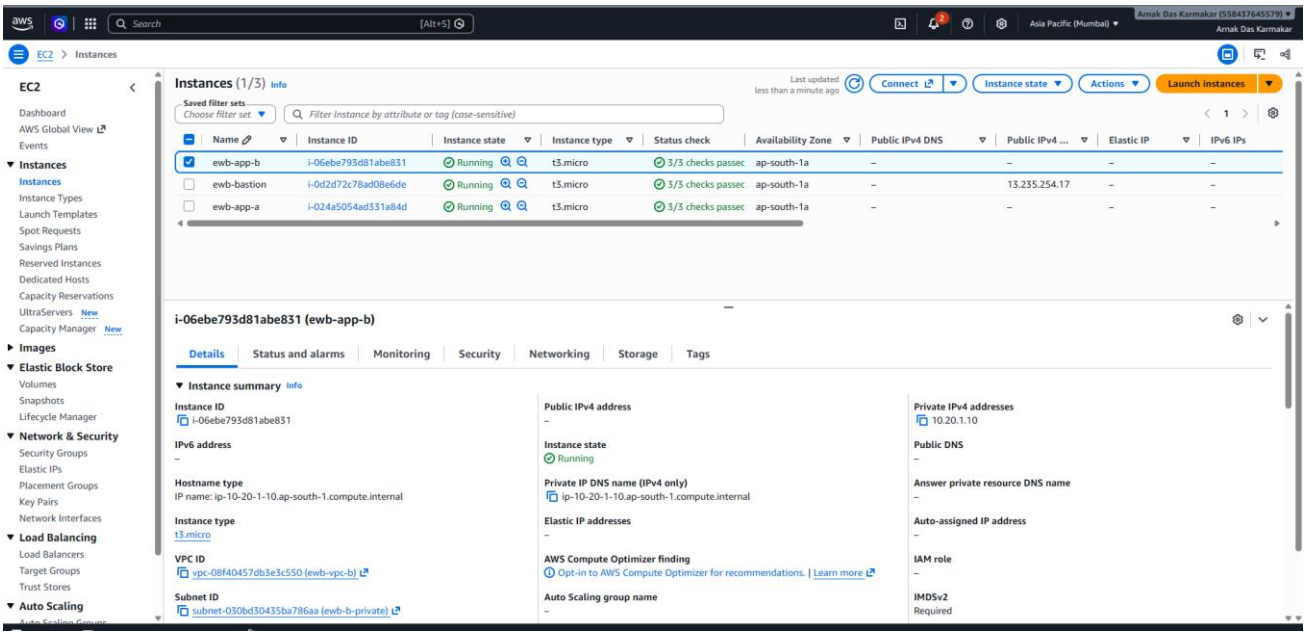
8. Cleanup and evidence checklist

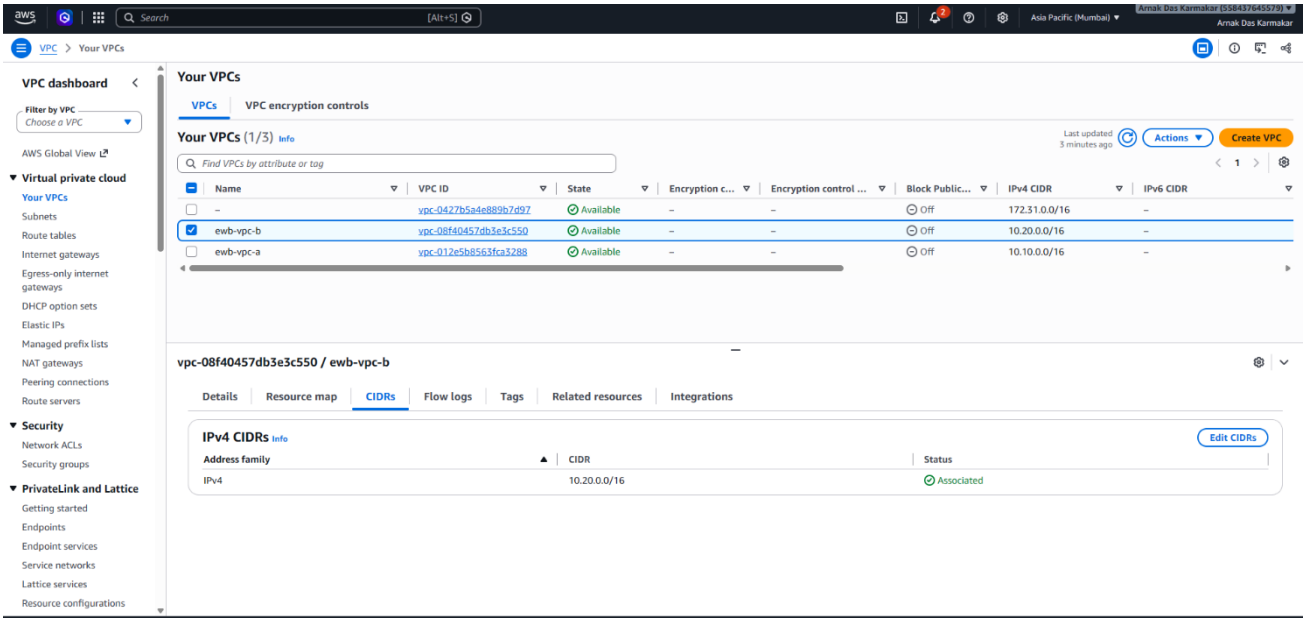
The workbook requires saving evidence before removing lab resources. Cleanup completion was not confirmed in the conversation, so mark each item only after checking it in the AWS Console.

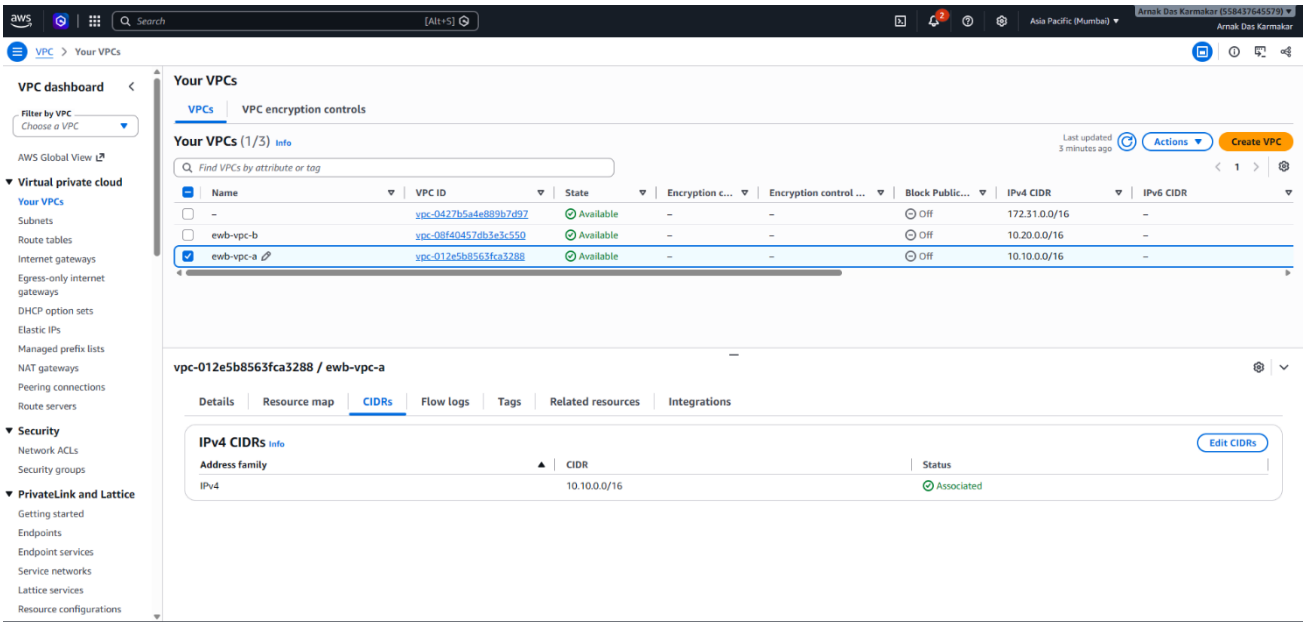
Cleanup item	Verify / record	Complete
Terminate EC2 instances	ewb-bastion, ewb-app-a, ewb-app-b terminated	■
Storage and public IPs	No leftover lab EBS volumes/snapshots or Elastic IPs	■
Peering	Peer routes removed; ewb-a-b-peer deleted	■
Network resources	Lab security groups, subnets and custom route tables deleted	■
VPC dependencies	ewb-igw-a detached/deleted; both lab VPCs deleted	■
Final audit	Region checked for leftovers; Billing reviewed; cleanup screenshots saved	■

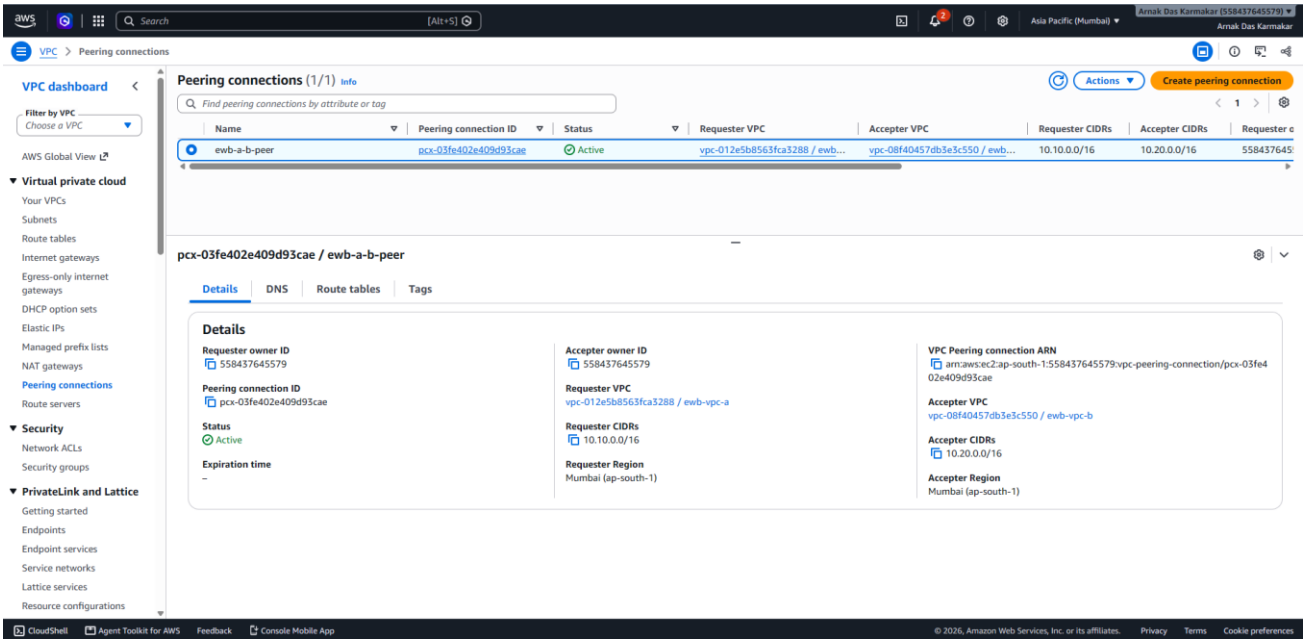

Instances (1/3)
ewb-app-b i-06ebe793d81abe831 Running t3.micro 3/3 checks passed ap-south-1a
ewb-bastion i-0d2d72c78ad08e6de Running t3.micro 3/3 checks passed ap-south-1a 13.235.254.17
ewb-app-a i-024a5054ad331a84d Running t3.micro 3/3 checks passed ap-south-1a
i-024a5054ad331a84d (ewb-app-a)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-024a5054ad331a84d
Public IPv4 address
Private IPv4 addresses
10.10.2.10
Instance state
Running
Public DNS
Hostname type
IP name: ip-10-10-2-10.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-10-2-10.ap-south-1.compute.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
VPC ID
vpc-012e5b8563fca3288 (ewb-vpc-a)
AWS Compute Optimizer finding
IAM role
Subnet ID
subnet-0bb0420f08b4793f2
Auto Scaling group name
IMDSv2
Required

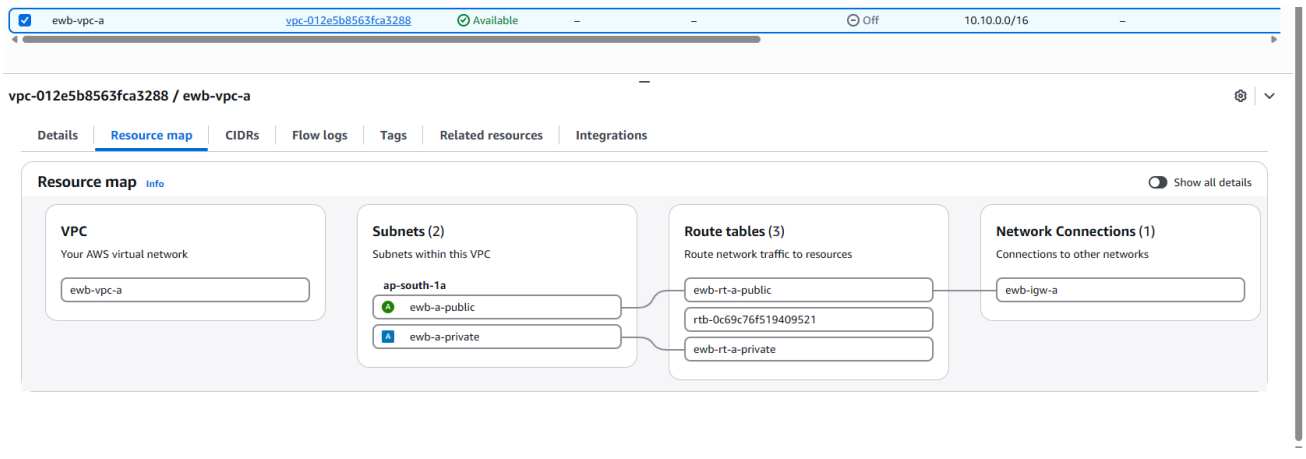

Instances (1/3)
ewb-app-b i-06ebe793d81abe831 Running t3.micro 3/3 checks passed ap-south-1a
ewb-bastion i-0d2d72c78ad08e6de Running t3.micro 3/3 checks passed ap-south-1a 13.235.254.17
ewb-app-a i-024a5054ad331a84d Running t3.micro 3/3 checks passed ap-south-1a
i-0d2d72c78ad08e6de (ewb-bastion)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-0d2d72c78ad08e6de
Public IPv4 address
13.235.254.17 | open address
Private IPv4 addresses
10.10.1.10
IPv6 address
Instance state
Running
Public DNS
Hostname type
IP name: ip-10-10-1-10.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-10-1-10.ap-south-1.compute.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
VPC ID
vpc-012e5b8563fca3288
AWS Compute Optimizer finding
IAM role
Subnet ID
subnet-04f4b62325d2f2e3c
Auto Scaling group name
IMDSv2
Required

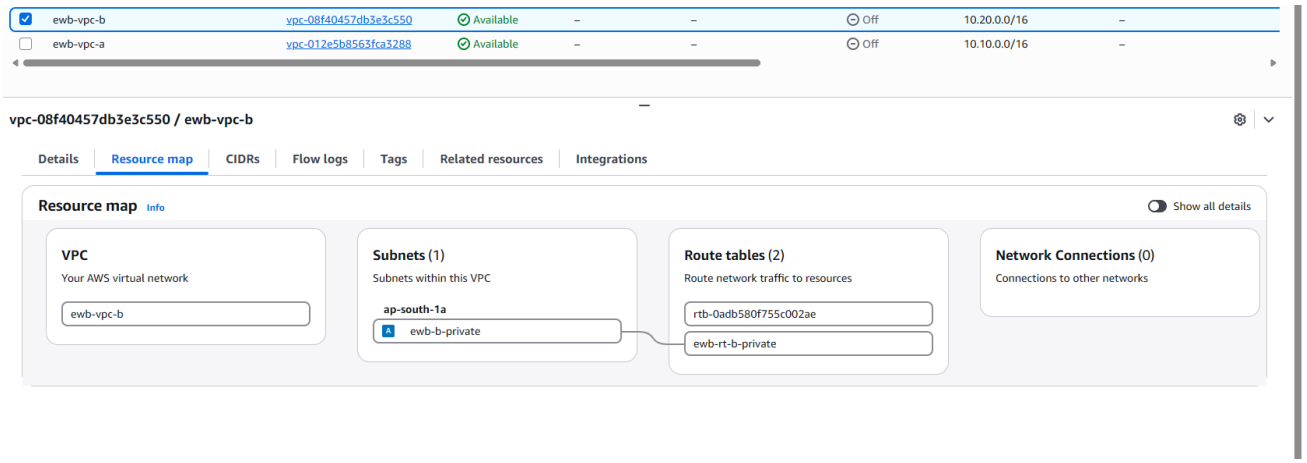

EC2 > Instances
Instances (1/3) Info
Launch instances
ewb-app-b i-06ebe793d81abe831 Running t3.micro 3/3 checks passed ap-south-1a
ewb-bastion i-0d2d72c78ad08e6de Running t3.micro 3/3 checks passed ap-south-1a 13.235.254.17
ewb-app-a i-024a5054ad331a84d Running t3.micro 3/3 checks passed ap-south-1a
i-06ebe793d81abe831 (ewb-app-b)
Details Status and alarms Monitoring Security Networking Storage Tags
Instance summary Info
Instance ID i-06ebe793d81abe831
Public IPv4 address –
Private IPv4 addresses 10.20.1.10
IPv6 address –
Instance state Running
Public DNS –
Hostname type IP name: ip-10-20-1-10.ap-south-1.compute.internal
Private IP DNS name (IPv4 only) ip-10-20-1-10.ap-south-1.compute.internal
Answer private resource DNS name –
Instance type t3.micro
Elastic IP addresses –
Auto-assigned IP address –
VPC ID vpc-08f40457db3e3c550 (ewb-vpc-b)
AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role –
Subnet ID subnet-030bd30435ba786aa (ewb-b-private)
Auto Scaling group name –
IMDSv2 Required

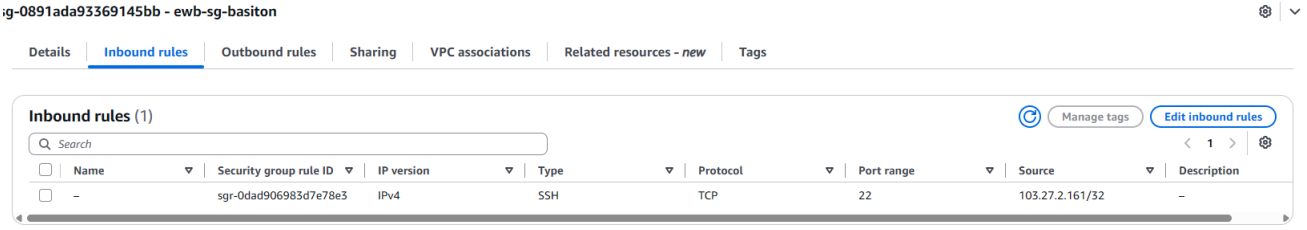

VPC > Your VPCs
Your VPCs
VPCs VPC encryption controls
Your VPCs (1/3) Info
Create VPC
ewb-vpc-b vpc-08f40457db3e3c550 Available Off 10.20.0.0/16
ewb-vpc-a vpc-012e5b8563fca3288 Available Off 10.10.0.0/16
vpc-08f40457db3e3c550 / ewb-vpc-b
Details Resource map CIDRs Flow logs Tags Related resources Integrations
IPv4 CIDRs Info
Edit CIDRs
Address family CIDR Status
IPv4 10.20.0.0/16 Associated

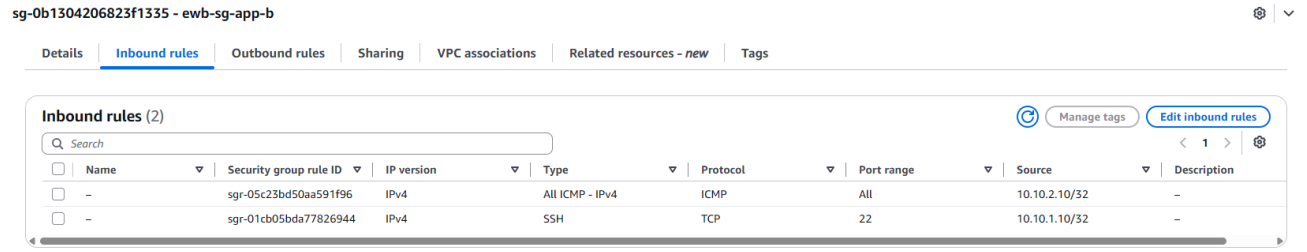

VPC dashboard
Your VPCs
VPCs
VPC encryption controls
Your VPCs (1/3) Info
Create VPC
Actions
ewb-vpc-b
ewb-vpc-a
vpc-012e5b8563fca3288 / ewb-vpc-a
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
IPv4 CIDRs Info
Edit CIDRs
Address family
CIDR
Status
IPv4
10.10.0.0/16
Associated

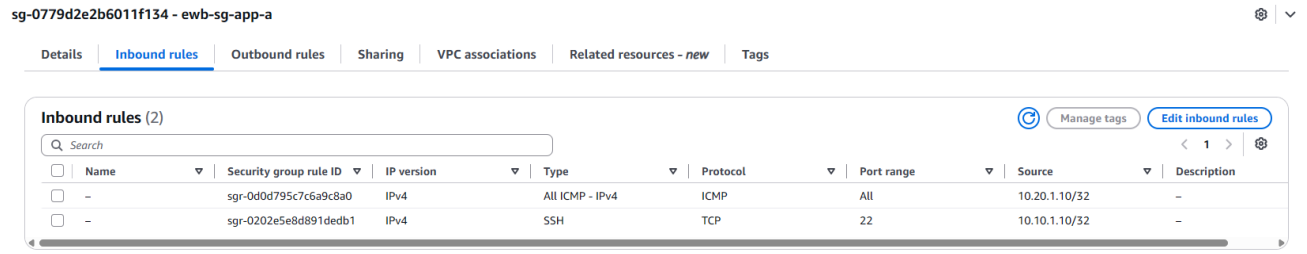

VPC dashboard
Peering connections
Peering connections (1/1) Info
Create peering connection
Actions
ewb-a-b-peer
pcx-03fe402e409d93cae
Active
pcx-03fe402e409d93cae / ewb-a-b-peer
Details
DNS
Route tables
Tags
Requester owner ID
558437645579
Peering connection ID
pcx-03fe402e409d93cae
Status
Active
Expiration time
Accepter owner ID
Requester VPC
vpc-012e5b8563fca3288 / ewb-vpc-a
Requester CIDRs
10.10.0.0/16
Requester Region
Mumbai (ap-south-1)
VPC Peering connection ARN
Accepter VPC
vpc-08f40457db3e3c550 / ewb-vpc-b
Accepter CIDRs
10.20.0.0/16
Accepter Region
Mumbai (ap-south-1)

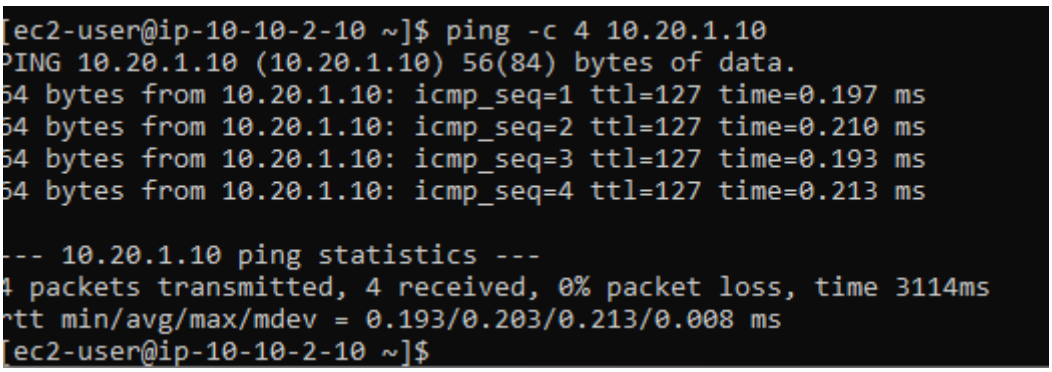

ewb-vpc-a
vpc-012e5b8563fca3288
Available
Off
10.10.0.0/16
vpc-012e5b8563fca3288 / ewb-vpc-a
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
ewb-vpc-a
Subnets (2)
Subnets within this VPC
ap-south-1a
ewb-a-public
ewb-a-private
Route tables (3)
Route network traffic to resources
ewb-rt-a-public
rtb-0c69c76f519409521
ewb-rt-a-private
Network Connections (1)
Connections to other networks
ewb-igw-a


ewb-vpc-b
vpc-08f40457db3e3c550
Available
Off
10.20.0.0/16
ewb-vpc-a
vpc-012e5b8563fca3288
Available
Off
10.10.0.0/16
vpc-08f40457db3e3c550 / ewb-vpc-b
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
ewb-vpc-b
Subnets (1)
Subnets within this VPC
ap-south-1a
ewb-b-private
Route tables (2)
Route network traffic to resources
rtb-0adb580f755c002ae
ewb-rt-b-private
Network Connections (0)
Connections to other networks


g-0891ada93369145bb - ewb-sg-basiton
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
–
sgr-0dad906983d7e78e3
IPv4
SSH
TCP
22
103.27.2.161/32
–


sg-0b1304206823f1335 - ewb-sg-app-b
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
–
sgr-05c23bd50aa591f96
IPv4
All ICMP - IPv4
ICMP
All
10.10.2.10/32
–
–
sgr-01cb05bda77826944
IPv4
SSH
TCP
22
10.10.1.10/32
–


sg-0779d2e2b6011f134 - ewb-sg-app-a
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
Description
–
sgr-0d0d795c7c6a9c8a0
IPv4
All ICMP - IPv4
ICMP
All
10.20.1.10/32
–
–
sgr-0202e5e8d891dedb1
IPv4
SSH
TCP
22
10.10.1.10/32
–

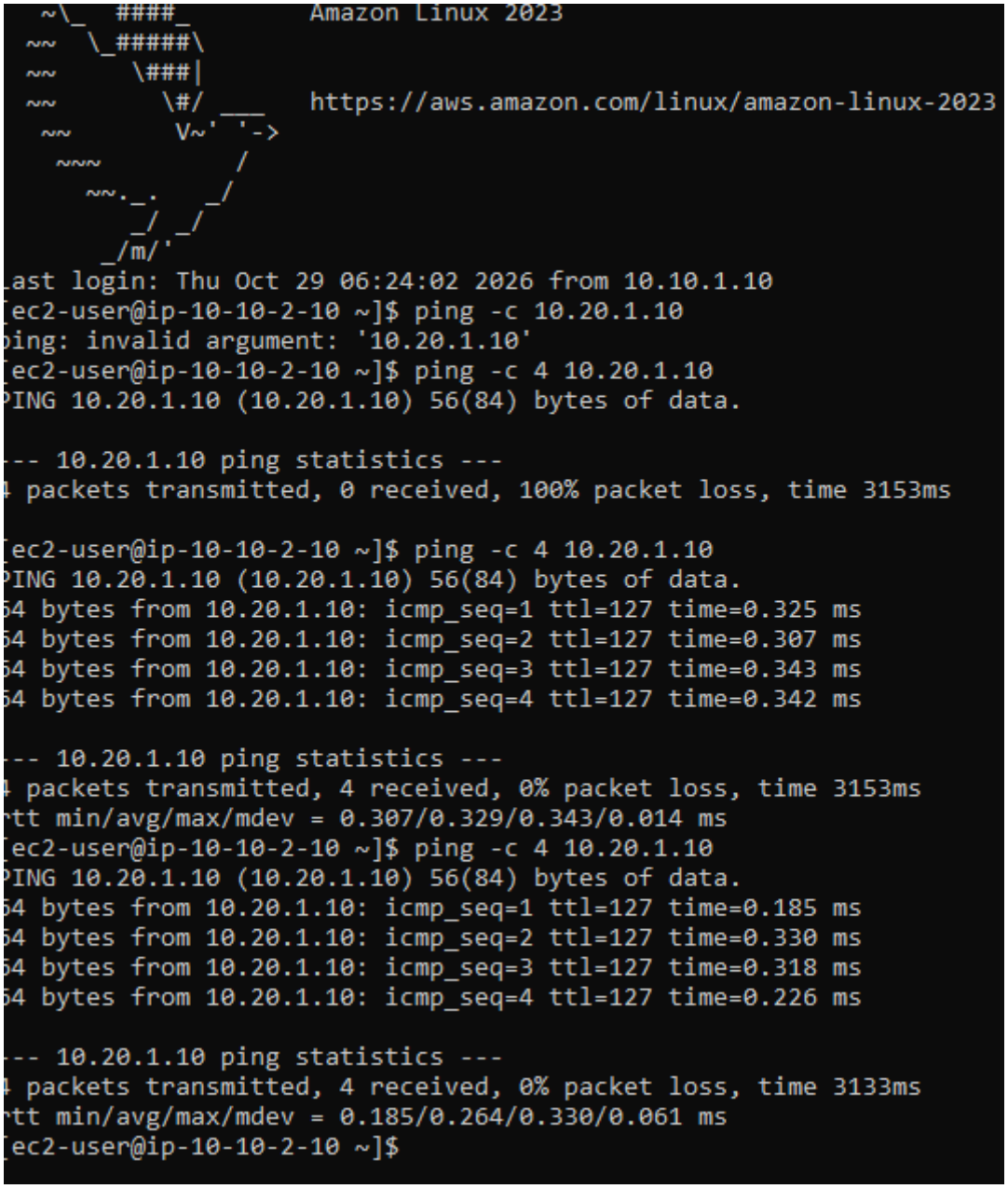

ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.197 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.210 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.193 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.213 ms
-- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3114ms
rtt min/avg/max/mdev = 0.193/0.203/0.213/0.008 ms
ec2-user@ip-10-10-2-10 ~]$

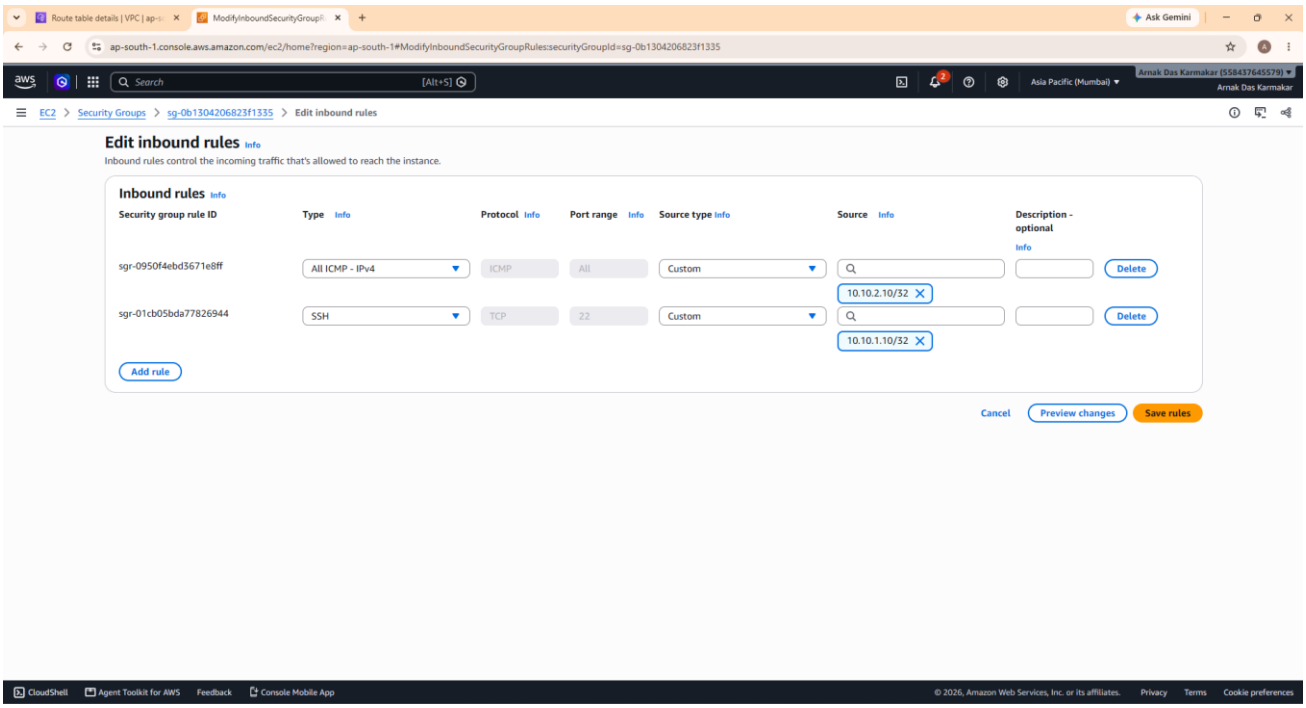
```
PS C:\WINDOWS\system32> ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 06:33:28 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3096ms
```

```
PS C:\WINDOWS\system32> ssh ewb-app-b
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 29 06:28:59 2026 from 10.10.1.10
[ec2-user@ip-10-20-1-10 ~]$ hostname
ip-10-20-1-10.ap-south-1.compute.internal
[ec2-user@ip-10-20-1-10 ~]$ hostname -I
10.20.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.234 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.222 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.216 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.236 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3088ms
rtt min/avg/max/mdev = 0.216/0.227/0.236/0.008 ms
[ec2-user@ip-10-20-1-10 ~]$
```


Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
ast login: Thu Oct 29 06:24:02 2026 from 10.10.1.10
ec2-user@ip-10-10-2-10 ~]$ ping -c 10.20.1.10
ing: invalid argument: '10.20.1.10'
ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
ING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
-- 10.20.1.10 ping statistics ---
packets transmitted, 0 received, 100% packet loss, time 3153ms
ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
ING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
4 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.325 ms
4 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.307 ms
4 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.343 ms
4 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.342 ms
-- 10.20.1.10 ping statistics ---
packets transmitted, 4 received, 0% packet loss, time 3153ms
tt min/avg/max/mdev = 0.307/0.329/0.343/0.014 ms
ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
ING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
4 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.185 ms
4 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.330 ms
4 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.318 ms
4 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.226 ms
-- 10.20.1.10 ping statistics ---
packets transmitted, 4 received, 0% packet loss, time 3133ms
tt min/avg/max/mdev = 0.185/0.264/0.330/0.061 ms
ec2-user@ip-10-10-2-10 ~]$

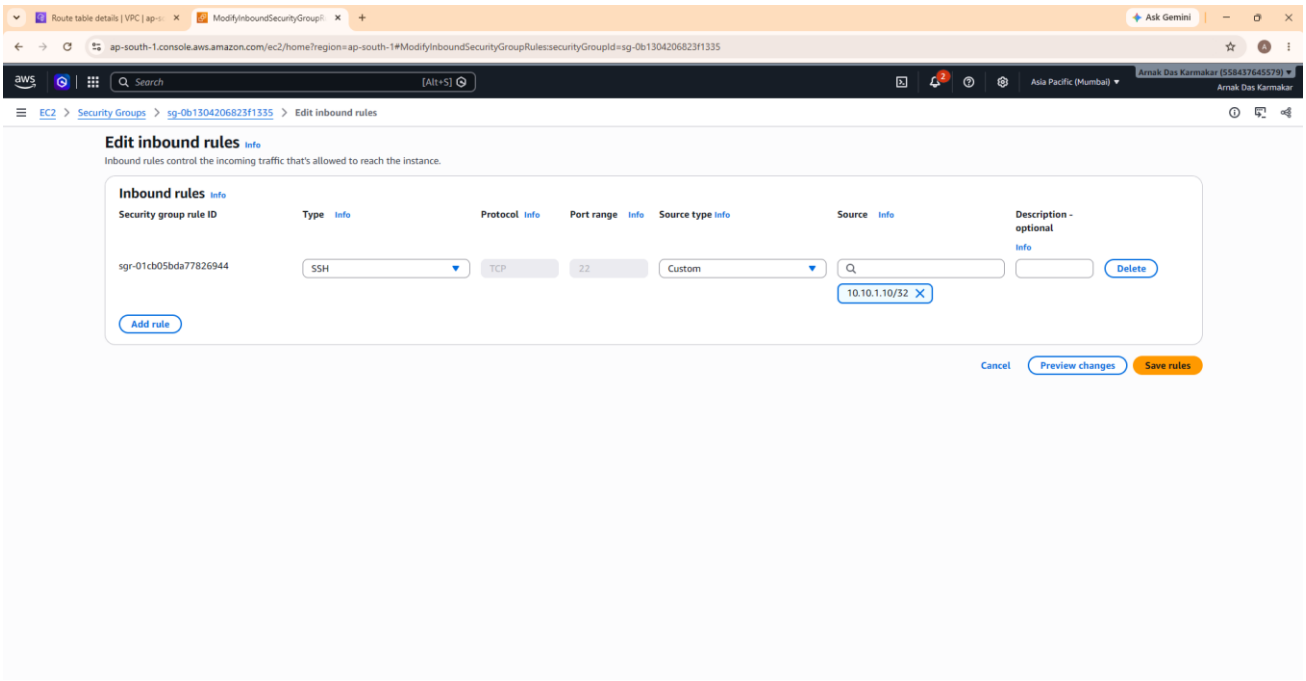


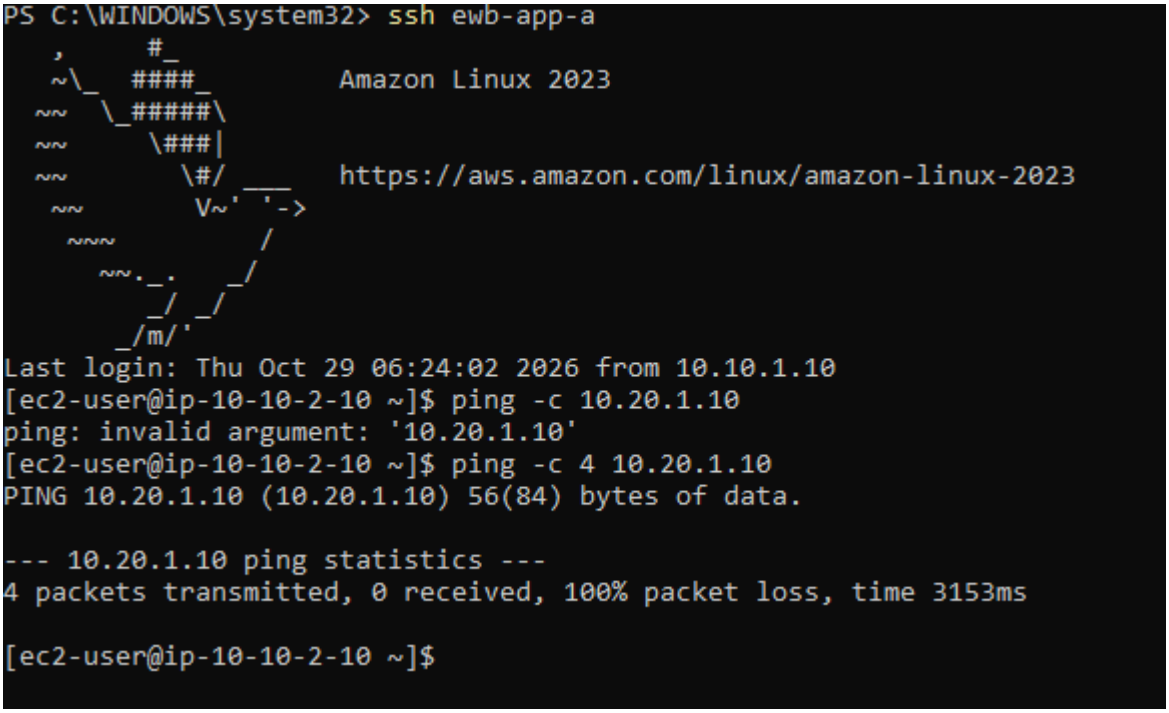

```
S C:\WINDOWS\system32> ssh ewb-app-a
       #_
       ####_        Amazon Linux 2023
      _#####\
      \###|
       \#/ ___     https://aws.amazon.com/linux/amazon-linux-2023
        V~' '->
         /
       _/
      _/ _/
    _/m/'
ast login: Thu Oct 29 06:24:02 2026 from 10.10.1.10
ec2-user@ip-10-10-2-10 ~]$ ping -c 10.20.1.10
ing: invalid argument: '10.20.1.10'
ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
ING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

-- 10.20.1.10 ping statistics ---
 packets transmitted, 0 received, 100% packet loss, time 3153ms

ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
ING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
4 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.325 ms
4 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.307 ms
4 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.343 ms
4 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.342 ms

-- 10.20.1.10 ping statistics ---
 packets transmitted, 4 received, 0% packet loss, time 3153ms
tt min/avg/max/mdev = 0.307/0.329/0.343/0.014 ms
ec2-user@ip-10-10-2-10 ~]$
```





Conclusion and learning outcomes

The project applied VPC CIDR planning, subnet separation, route-table association, internet-gateway configuration, VPC peering, stateful security-group rules and bastion-based administration. The two application hosts communicated over private IPv4 addresses once both forward and return routes and the required ICMP rules were in place. The controlled failure/recovery exercise showed how a security-group rule can intentionally block and then restore traffic. Final submission should include the student's own screenshots, actual AWS resource IDs and verified cleanup status.

9. Student details and actual resource IDs

VPC A ID	vpc-012e5b8563fca3288	VPC B ID	vpc -08f40457db3e3c550
Public subnet ID	subnet-04f4b62325d2f2e3c	Private A subnet ID	subnet-0bb0420f08b4793f2
Private B subnet ID	subnet-030bd30435ba786aa	Peering ID	pcx-03fe402e409d93cae
Bastion instance ID	i-0d2d72c78ad08e6de	App-A instance ID	i-024a5054ad331a84d
VPC A ID	vpc-012e5b8563fca3288	VPC B ID	vpc-08f40457db3e3c550
App-B instance ID	i-06ebe793d81abe831	Cleanup verified on	_____

Security note: do not include private keys, passwords, access tokens, or other credentials in the report.